

LUCAS MARTINEZ

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EDUCATION

New York University

Master's, Computer Science

September 2023 - May 2025

GPA: 3.88

- Relevant Coursework: Machine Learning, Predictive Analytics, Computer Vision, Deep Learning, ML and Cloud

Universidad Catolica Nuestra Señora de la Asuncion

Bachelor's, Computer Science

January 2017 - November 2022

SKILLS

Technical Skills: Machine Learning, Deep Learning, LLM, Vector Databases, RAG, Pytorch, Tensorflow, Forecasting, Regression, Embeddings, Clustering, End-To-End

Programming Languages: Python, C++, Go, JavaScript, SQL, NoSQL

Soft Skills: Problem-Solving, Curious, Creative, Cross-Functional, Attention to Detail, Fast Learner, Agile

PROFESSIONAL EXPERIENCE

CANARIA

Machine Learning Engineer (Intern)

New York, NY, USA

May 2025 - Present

- Built and optimized a large-scale deduplication pipeline for job postings using Milvus, MongoDB, and Union-Find, processing over 400 million job records.
- Designed a vector similarity search system, achieving efficient duplicate detection across hundreds of millions of text embeddings.
- Automated the ingestion, comparison, and clustering process with Python, Docker and Kafka, enabling scalable daily insertion of 100,000+ new jobs data.

TECHNOLOGIES DEVELOPMENT OF PARAGUAY (TDP)

Software Engineer

Asunción, Paraguay

January 2022 - August 2023

- Managed and optimized in-server payment processing for 400,000+ users, handling over 3 million daily transactions and reducing processing time by 80%
- Led the development of a scalable digital banking and wallet system, serving 60,000+ users, with AI-driven user verification (computer vision, facial recognition, OCR integration)
- Directed the initial steps of an LLM chatbot implementation, establishing a scalable document processing and retrieval system.
- Enhanced the in-app public transit routing system with real-time data, improving accuracy and boosting user satisfaction (gained viral traction on Twitter).

CADI S.A.

Software Developer

Asunción, Paraguay

November 2018 - February 2020

- Built a robust billing system for a web platform serving small and medium businesses, streamlining operations and reducing manual errors
- Redesigned the inventory control system enabling real-time inventory updates across multiple devices.

PROJECTS & OUTSIDE EXPERIENCE

Fantastic Router: an open-source LLM-powered intent router

Python, LLM, FastAPI, Docker

New York, NY, USA

August 2025 - Present

- Developed a natural language routing engine enabling users to navigate complex applications via plain queries like “show me James Smith’s monthly income”, mapping directly to routes such as `/landlords/jamessmith123/financials`
- Designed with extensibility in mind, supporting pluggable adapters for databases, LLM providers, and vector stores, along with built-in rolebased access control (RBAC)
- [Link to project](#)

Propi - Property Management & Financial System

JavaScript, NodeJS, React, Supabase

June 2025 - Present

- A full-featured React/TypeScript SaaS platform for property management, built with a Node.js backend
- It includes automated invoice processing, AI-powered document analysis using Mistral OCR and Gemini AI, real-time payment tracking, and integrated WhatsApp notifications
- Supports multi-currency operations, handles tax calculations, and automates tenant communication workflows end to end.
- [Link to project](#)

Duplicated Job Posting Detection System

New York, NY, USA

Python, Milvus DB, Docker, Embeddings

April 2025 - Present

- Developed a job duplicate detection system using SentenceTransformers and Milvus for semantic similarity search
- Also integrated a FastAPI service for real-time matching and evaluation with custom threshold tuning.
- [Link to project](#)

JobRAG

New York, NY, USA

Python, LLM, Qdrant, Embeddings

April 2025 - Present

- Designed a system that combines retrieval-augmented generation with job embeddings to enhance job search and exploration
- Integrated Qdrant for vector search and FastAPI for dynamic query handling and stats visualization.
- [Link to project](#)

The Backprop Blog: a Machine Learning Blog

New York, NY, USA

Developer & Author - TypeScript, JavaScript

March 2025 - Present

- Designed and developed a personal technical blog using Astro, Markdown, and Vercel, where I publish in-depth posts on machine learning, computer vision, and deep learning
- I regularly post in-depth articles that combine theoretical concepts with practical implementations, covering topics like CNN, attention mechanisms, and model optimization
- The blog emphasizes clarity, visual explanations, and technical accuracy to support both learning and knowledge sharing.
- [Link to project](#)

GradRag LLM

New York, NY, USA

Python, Qdrant DB, Embeddings

November 2024 - Present

- A Retrieval-Augmented Generation (RAG) application for PhD aspirants to search professors/tutors using embeddings and vector-based similarity
- Implemented and compared multiple NLP models, including TF-IDF, BERT, and SBERT, for optimized search accuracy.

Hedge Fund alpha: Clustering stocks for portfolio enhancement using a novel approach

New York, NY, USA

Python, Data Mining, Data Analysis, Clustering

September 2024 - December 2024

- Implemented a predictive model to classify sector-based stock clusters, leveraging alternative data sources
- Analyzed interdependencies within S&P 500 indices and 13-F filings, using a new proposed technique, to inform strategic portfolio decisions, outperforming the S&P 500 (10.22% vs 6.5%)

Mentorship: Mentor in Software Engineering

Remote

Python, C++, Docker, Kubernetes

2022 - 2022

- Co-mentored alongside an NVIDIA Autonomous Car Machine Learning Infrastructure Engineer, helping students build and deploy microservice applications.

IBM DATAJAM "PASOS LIBRES": ForcedData

Remote

Python, JavaScript, Data Science, IBM Cloud

2022 - 2022

- Led a team of four to develop ForcedData, a data-driven platform to help users identify and act on potential forced labor situations using machine learning, big data, and the cloud.

PUBLICATIONS

- **On the Measurement of io_uring Performance: a Strategy and Experience in the Envoy Service Mesh -**
<https://doi.org/10.19153/cleiej.27.2.3>
- **A Measurement Strategy for io_uring Performance in the Envoy Service Mesh -**
<https://doi.org/10.1109/CLEI60451.2023.10346093>